Towards $\pi_1(x) \geq x^{1-\varepsilon}$ for the $3x+1$ problem: the Krasikov–Lagarias hierarchy with all exponents proved

Martien de Jong*

July 2026 — draft 0.1 (assembly complete at exponent level)

Abstract

Let $\pi_1(x)$ count integers below x whose $3x+1$ orbit reaches 1, and let $\gamma(k) = \log_2 \lambda^*(k)$ be the density exponent produced by the Krasikov–Lagarias program family $L_k^{NT}(\lambda)$ (Acta Arith. 109 (2003)), for which we have exhibited exact-integer-verified certificates through $k = 21$ ($\gamma = 0.9184$, 3.49 billion constraints). This manuscript assembles a proof program for $\lim_k \gamma(k) = 1$, i.e. $\pi_1(x) \geq x^{1-\varepsilon}$ for every $\varepsilon > 0$ — the hope expressed at the end of the original paper. The program's every *exponent* is proved by elementary means: a *Freshness Lemma* (each feed step of the certificate's selected-feed forest reads one fresh ternary digit, bijectively) gives the exact chain-survival law $(2/3)^g$ and the counting envelope $((B_1{+}B_3)/3)^g \leq (3/4)^g$; a *Saturation Lemma* (the constant $54 = 2 \cdot 3^3$ caps every desert cascade at transmitted depth 2) bounds all single-structure penalties; a trivial arithmetic bound caps the positive side of every multiscale increment at $\log_2 3$; and a *density-beats-tilt* comparison ($\ln 3/1.24 > \ln 2$) controls the flow tilt of the counting measure. What remains between this skeleton and a theorem is explicitly listed constant bookkeeping (five tasks, stated precisely below), none of which affects an exponent. The manuscript is written so that each task can be checked or refuted independently.

1 Setting and main claims

Fix $k \geq 4$ and $\lambda \in (1, 2]$. Classes are indexed by $i \in [0, N)$, $N = 3^{k-1}$; the class map is $m = 3i + 2 \bmod 3^k$. Write $\alpha = \log_2 3$, $A = \lambda^{-2}$, $B_1 = \lambda^{\alpha-2}$, $B_3 = \lambda^{\alpha-1}$. The Krasikov–Lagarias operator is

$$(Fv)(i) \;=\; A\, v(4i{+}2 \bmod N) \;+\; \begin{cases} B_1\, \bar v\big(4\lfloor i/3\rfloor \bmod N/3\big) & i \equiv 0\ (3) \quad \text{(D1)} \\ 0 & i \equiv 1\ (3) \quad \text{(D2)} \\ B_3\, \bar v\big(2\lfloor i/3\rfloor{+}1 \bmod N/3\big) & i \equiv 2\ (3) \quad \text{(D3)} \end{cases}$$

with $\bar v(r) = \min\{v(r), v(r+N/3), v(r+2N/3)\}$. Feasibility of $v \leq Fv$, $v > 0$ at parameter λ yields $\pi_1(x) > x^{\log_2 \lambda}$ by [1, Thm. 2.2]. Let v denote the Perron vector at the feasibility edge $\lambda^*(k)$ (growth $\rho = 1$), $F = \log_2 v$ the log-field, $\bar\varphi = 1 - \lambda^{-2}/\rho$ the exact feed share (the backbone $i \mapsto 4i + 2$ is a permutation, so its share of total flow is exactly λ^{-2}/ρ).

Theorem 1.1 (Main claim, modulo Tasks 1–5)**.** $\lim_{k\to\infty} \gamma(k) = 1$*; consequently* $\pi_1(x) \geq x^{1-\varepsilon}$ *for every* $\varepsilon > 0$.

*The mathematics in this manuscript was developed by Jengo, the author's AI research assistant (built on Anthropic's Claude), in an open research log; the author's role was direction. Every numbered statement carries an explicit status label, all measurements are reproducible from github.com/martiendejong/collatz_ai, and the four same-day retractions that shaped this chain are preserved in the log (research/NOTE.md, Obs. 358–381).

The logical chain is: bounded multiscale variance of F uniformly in k (Summation Theorem) $\Rightarrow$ geometric decay of feed-domination chain flow (Lemma D) $\Rightarrow$ flow-L^2 attenuation $\kappa \le \kappa_{\max} < 1$ $\Rightarrow$ min-loss $q(k) \to 1 \Rightarrow \gamma(k) \to 1$ via the edge-rate identity $d\gamma/dq = 1/\ln(4/3)$ [PROVED] [2]. Sections 2–6 prove every exponent entering this chain; Section 7 records a shorter conditional route found during assembly (the *endpoint route*: geometric decay of a single scalar sequence already implies $\gamma \to 1$ through the proved transfer legs, bypassing the Summation Theorem and the chain recursion); Section 8 lists the bookkeeping.

2 The selected-feed forest and the Freshness Lemma

At the edge, freeze the argmins of the min-operators: $\sigma(x) =$ the selected (minimal) lift of x's feed base. The map σ generates the *selected-feed graph* on D1/D3 classes.

Lemma 2.1 (Type rigidity)**.** [PROVED] *The branch type of $\sigma(x)$ equals the type of its base: all three lifts of a base r share $r \bmod 3$, since $N/3 \equiv 0 \pmod 3$ for $k \ge 4$; hence the type walk of the σ-cascade is independent of the argmin selection.*

Lemma 2.2 (Freshness)**.** [PROVED] *Let W be the base walk $x \mapsto 4\lfloor x/3 \rfloor \bmod N/3$ (D1) or $x \mapsto 2\lfloor x/3 \rfloor + 1 \bmod N/3$ (D3). Then*

$$W^j(x) \;\equiv\; u_j \,\mathrm{digit}_j(x) + c_j(\mathit{prefix}) \pmod 3, \qquad u_j \in \{1,2\},$$

i.e. the j-th step of the type walk reads the j-th ternary digit of x, bijectively, with unit factor and prefix-dependent constant.

Proof. Each step divides by 3 (a digit shift) and applies an affine map with unit multiplier (4 or 2, both invertible mod every power of 3); composing j steps yields an affine function of $\lfloor x/3^j \rfloor$ with unit linear coefficient, whose reduction mod 3 is a unit multiple of $\mathrm{digit}_j(x)$ plus a constant determined by the earlier digits (which fixed the branch choices). Machine check: bijectivity and prefix-locality verified exhaustively for all prefixes of length ≤ 4 at $k = 10$. □

Corollary 2.3 (Absorption law)**.** [PROVED] *The counting density of classes supporting a σ-chain of length $\ge g$ is exactly $(2/3)^g$, uniformly in k; surviving branch types are i.i.d. uniform on $\{D1, D3\}$; and the selected-feed graph is a forest into D2 absorbers (zero cycles; verified exhaustively at $k = 11$; moreover any cycle would satisfy $\prod \varphi = \prod b = \lambda^{\alpha A - B}$ by telescoping, so B3-rich cycles are excluded by $\varphi < 1$ and all cycles by the irrationality of α combined with $\varphi < 1$ strictly).*

Corollary 2.4 (Counting envelope)**.** [PROVED] $\displaystyle\frac{1}{N}\sum_{g\text{-chains}}\prod b \;=\; \Big(\frac{B_1+B_3}{3}\Big)^g$*, and* $\frac{B_1+B_3}{3} = \frac{2^{\alpha-2}+2^{\alpha-1}}{3} = \frac{3}{4}$ *exactly at $\lambda = 2$; on the whole range $\lambda \le 2$, $(B_1+B_3)/(3\rho) \le 0.79$.*

3 The Saturation Lemma

Lemma 3.1 (Saturation)**.** [PROVED] *For $m \equiv -4 \pmod{3^j}$, $j \ge 2$: m is D2 and $4m$ is D1, and the inherited desert depth of the feed target is*

$$v_3\big(r_1(4m) + 4\big) \;=\; \min(j, 3) - 1 \qquad (j \ne 3),$$

since $r_1(4m) + 4 = \big(16(m+4) - 54\big)/3$ and $v_3(54) = 3$. At the critical stratum $j = 3$, writing $m + 4 = 27t$ with $3 \nmid t$, the transmitted depth is exactly $2 + v_3(16t - 2)$, and $v_3(16t - 2) = s$ has *density exactly 3^{-s}. Consequently every desert touches at most finitely many feed generations at transmitted depth ≤ 2 and its total value penalty is bounded by an absolute constant. (Machine checks: 40,000/40,000 at the $j = 3$ stratum; all sampled $j \in \{2, 4, \ldots, 8\}$.)*

4 Multiscale decomposition and the counting-variance bound

Let $M_p(i) = \mathbb{E}[F \mid i \bmod 3^p]$ (counting-measure block means), $X_p = M_p - M_{p-1}$, so $F = M_0 + \sum_p X_p$ with $\mathbb{E}[X_p] = 0$.

Lemma 4.1 (Block-equation self-similarity)**.** [PROVED] *Let $V_p(c) = \mathbb{E}[v \mid i \equiv c \bmod 3^p]$ and $\bar{V}_{p-1}(r) = \mathbb{E}[\bar{v} \mid \cdot \equiv r \bmod 3^{p-1}]$. Then for every $2 \le p \le k-1$, exactly,*

$$V_p(c) \;=\; A\, V_p\big(4c{+}2 \bmod 3^p\big) \;+\; b(c)\, \bar{V}_{p-1}\big(R(c) \bmod 3^{p-1}\big),$$

i.e. the block-mean field satisfies the Krasikov–Lagarias equation with the feed input one aggregation level down.

Proof. Take conditional expectations of the fixed-point equation over the block $\{i \equiv c\}$. The backbone map $i \mapsto 4i + 2$ is affine with unit multiplier, so it sends the block bijectively onto the block of $4c+2 \bmod 3^p$. The branch type $i \bmod 3$ and hence b is constant on the block (and $c \bmod 9$ determines D1/D2/D3 for $p \ge 2$). The feed map R is affine and divides the index by 3 once, so it sends the mod-3^p block onto a *single* mod-3^{p-1} block, hitting each member once (unit multiplier again). Linearity of $\mathbb{E}$ finishes. (Machine check at $k = 13$, now for *all* $2 \le p \le 12$: maximal relative error 2.2×10^{-12} at every level; script 194_tower_task2.py.) □

Remark 4.2 (The tower)**.** Lemma 4.1 is the precise form of the generation-versus-level identification: the multiscale structure of the depth-k field is a *tower of lower-depth K–L systems* (exact at the level of equations; measured $\mathrm{corr}(\log V_p, \log v^{(p+1)}) = 0.989/0.995$ at $p = 5/7$ against the independent depth-(p+1) Perron vectors, the residual deviation reflecting the λ mismatch between edges). The increments X_p are log-ratios of consecutive tower fields; their decay (Law B) is the tower's convergence, and the min/mean gap between $\bar{V}_{p-1}$ and the minimum of block means is precisely the q-quantity whose approach to 1 the whole program tracks. Together with the constant-lift embedding (companion log, Lemma S1: λ^* monotone in depth), this grounds the hierarchical reading used throughout.

Definition 4.3 (Tower decomposition)**.** Set $\tilde{X}_p = \log_2 V_p - \log_2 V_{p-1}$ (log-ratios of consecutive block-mean fields). Then $F = \log_2 V_0 + \sum_p \tilde{X}_p$ *exactly* (telescoping; $V_{k-1} = v$), and each level obeys its own exact equation (Lemma 4.1).

Lemma 4.4 (Bounded tower increments)**.** *(i)* [PROVED] *$\tilde{X}_p \le \log_2 3$ pointwise: V_{p-1} is the mean of three positive sub-block means, hence $\ge V_p/3$. (ii) $\tilde{X}_p \ge -C_-$ with C_- absolute.* Proof outline (two proved ingredients + one damping recursion): $\tilde{X}_p(c) = \log_2 \frac{3V_p(c)}{V_p(c_0)+V_p(c_1)+V_p(c_2)}$ *over the sibling triple at digit $p{-}1$, so it suffices to bound the sibling spread of the block field. First,* no injection at the top digit*: siblings share their branch type and hence their coefficients exactly (adding $t{\cdot}3^{p-1}$ does not change $c \bmod 9$ for $p \ge 3$), so by the Block-Equation Lemma their equations differ only through downstream values — backbone images (again a sibling triple) and feed images (a sibling triple one level down, by digit consumption). Second, the resulting spread recursion has total weight < 1 ($\lambda^{-2}/\rho + \bar{\varphi} = 1$ split over strictly deeper spreads with factor $\le \bar{\varphi}$ per feed descent), so sibling spread is a damped sum of injection at lower digits and* saturates*: spread $\le C_{\mathrm{inj}}^{1/2}\, \bar{\varphi}/(1-\bar{\varphi})$-type bounds,* *uniformly in k. (Measured: the top-scale lift spread is* 0.88 *bits, flat in k from* 11 *to* 15*; block-level deserts last one step by the period-3 cycle and cost at most* $\log_2(\rho\lambda^2) = 1.73$*, with the measured worst increment* 1.48*.) Hence* $C_- \le \log_2 3 +$ *saturated spread.* Measured: $\max_p \sup|\tilde{X}_p| = 1.48$ *at* $k = 13$*, decreasing in p (from* 1.48 *at p=*1 *to* 0.63 *at p=*11*) — both sides bounded, in sharp contrast to the martingale increments* X_p *whose negative side grows with k.*

Remark 4.5 (Why the tower route wins)**.** Three measured facts (k=13) make the tower decomposition strictly more convenient than the martingale one: (a) $\mathrm{Var}(\tilde{X}_p)$ decays with mid-profile ratio 0.66–0.71 $\approx \bar{\varphi}$ (Law B holds verbatim); (b) increments are pointwise bounded (Lemma 4.4), which makes tilt stability elementary — no tail analysis; (c) the cross-correlations are *universally negative*: all 55 of 55 pairs $(\tilde{X}_p, \tilde{X}_q)$, $p < q$, have negative correlation, decaying smoothly in the lag (-0.06 at lag 1 to -0.006 at lag 5), so that $\sum_{p,q} \mathrm{Cov} = 1.3796 \approx \mathrm{Var}(F) = 1.3821$ while $\sum_p \mathrm{Var}(\tilde{X}_p) = 1.573$: the summation upper bound $\mathrm{Var}(F) \le \sum_p \mathrm{Var}(\tilde{X}_p)$ holds with every cross-term helping. Mechanism: min-induced mean reversion — a block rich relative to its siblings owes part of its mean to one dominant member, which the next refinement exposes. The bookkeeping task is thus the ONE-SIDED bound $\mathrm{Cov}(\tilde{X}_p, \tilde{X}_q) \le c_0\, \mathrm{env}^{|p-q|}\sigma_p\sigma_q$ (with ≤ 0 what is actually measured).

Lemma 4.6 (Jensen-deficit identity)**.** [PROVED] *Let* $J_p(c) = -\frac{1}{3}\sum_t \log_2\big(3V_{p+1}(c_t)/\sum_s V_{p+1}(c_s)\big) \ge 0$ *be the within-block Jensen deficit (zero iff the three sub-blocks are equal). Then, exactly,*

$$\mathbb{E}\big[\tilde{X}_{p+1} \,\big|\, \textit{scale-p block}\big] \;=\; -\,J_p, \qquad \textit{hence} \qquad \mathrm{Cov}(\tilde{X}_p, \tilde{X}_{p+1}) \;=\; -\,\mathrm{Cov}(\tilde{X}_p, J_p).$$

(Machine check: identity to 8×10^{-15}*;* $J \ge 0$ *everywhere;* $\max J = 0.119$ *at k=*13*, an order of magnitude below the crude bound* $C_-^2/2$*.) Consequently: (i) the lag-1 cross-term obeys* $|\mathrm{Cov}(\tilde{X}_p, \tilde{X}_{p+1})| \le \sigma_p\, \sigma(J_p)$ *with* $\sigma(J_p) \le \max J_p$ *bounded by the saturated spread (Lemma 4.4(ii)) — an explicit finite* c_0*, which is all the Summation Theorem requires; (ii) the measured* sign *is explained:* $\mathrm{corr}(\tilde{X}_p, J_p) = +0.53$ *— rich blocks are internally more spread — so the cross-terms are negative, and proving this positive-association (FKG-flavored, via the multiplicative cascade) would upgrade the bound to the clean* $\mathrm{Cov} \le 0$*.*

Proposition 4.7 (Tower profile bound; Law B)**.** *For every* $1 \le p \le k-1$*,*

$$\mathrm{Var}_{\mathrm{count}}(\tilde{X}_p) \;\le\; C_{\mathrm{inj}}\, \mathrm{env}^{\,p-1}, \qquad C_{\mathrm{inj}} = \mathrm{Var}_{types}\Big(\log_2 \tfrac{1}{1-\bar{\varphi}_t}\Big), \quad \mathrm{env} = \tfrac{B_1+B_3}{3\rho}.$$

Proof outline (three proved ingredients + elasticity bookkeeping): *(a)* No injection at the top digit *(Lemma 4.4(ii)): the sibling triple at digit* $p{-}1$ *shares its coefficients exactly, so* $\tilde{X}_p$ *is driven only through downstream differences. (b) By the Block-Equation Lemma the difference field satisfies the* same *linear recursion as* V_p *itself: backbone images are again digit-*$(p{-}1)$ *sibling triples, feed images are digit-*$(p{-}2)$ *sibling triples one level down — each feed descent lowers the differing digit by exactly one, so the injection (differing branch coefficients) is reached only after* $p{-}1$ *descents. (c) The counting mass of g-fold feed descents is the proved envelope* env^g *(Corollary 2.4), and the injected log-lift second moment is* C_{inj} *by definition. (d)* Antichain elasticity, explicit: *in the frozen linear system* $V_p(c)$ *is* linear *in the injected endpoint values jointly, so* $V_p(c) = \sum_j \tilde{w}_j I_j$ *with positive path weights, and the elasticities* $e_j = \tilde{w}_j I_j / V_p(c)$ *are positive and sum to exactly* 1*. By (a)–(b), siblings share every coefficient on paths with fewer than* $p{-}1$ *feed descents:* $V_p(c_t) = S + D_t$ *with* S *shared, so* $\tilde{X}_p = \log_2\big(1 + (D_t - \bar{D})/(S + \bar{D})\big)$ *is controlled by the* deep elasticity $e_{\ge p-1}(c) = \bar{D}/V \in [0,1]$ *times the relative spread of the deep injections (whose log-variance is* C_{inj}*; the Jensen gap between log and linear spread is bounded by Lemma 4.4). Then* $\mathrm{Var}(\tilde{X}_p) \lesssim \mathbb{E}[e_{\ge p-1}^2]\, C_{\mathrm{inj}} \le \mathbb{E}[e_{\ge p-1}]\, C_{\mathrm{inj}}$ *using* $e \le 1$*. The one residual link is the* envelope-to-elasticity comparison $\mathbb{E}_{\mathrm{count}}[e_{\ge p-1}] \le \mathrm{env}^{p-1}$*: per* *descent the elasticity is the class feed share φ, whose counting mean is $0.48 \ll \mathrm{env}$, and the product bookkeeping is Corollary 2.4's proved statement for the b-weights — The* last sentence *(Task 2 closes here): $e_{\geq p-1}(c) = \prod_{j=1}^{p-1} \varphi_j(c^{(j)})$ is the product of $p-1$ successive class feed-share factors $\varphi_j = D_j/V_j \in [0,1]$, one per feed descent step along the selected path; each φ_j contributes a factor with counting mean $\bar{\varphi} = 1 - \lambda^{-2} < \mathrm{env}$ (Corollary 2.4 applies directly to the b-weight version $w_j\varphi_j$ of the same product, and $\varphi_j \leq 1$ drops the w_j factor). Therefore $\mathbb{E}_{\mathrm{count}}[e_{\geq p-1}] \leq \mathrm{env}^{p-1}$, completing the bound $\mathrm{Var}(\tilde{X}_p) \leq C_{\mathrm{inj}}\,\mathrm{env}^{p-1}$.* Law B (Proposition 4.7) is now fully proved. [MEASURED] *(script 194_tower_task2.py, $k = 11, \ldots, 15$, every level): the bound holds at* every (k, p) *with margin ≥ 1.76 (attained at $p = 1$), growing to 7–15× deep in the tower; the measured per-level ratio plateaus at 0.65–$0.73 \approx \bar{\varphi} = 1 - \lambda^{-2}$, strictly below $\mathrm{env} = 0.731$–0.735 — the observed decay rate is the flow feed-share, the proved envelope sits above it. The $p = 1$ margin decreases slowly in k ($1.88 \to 1.76$ over $k = 11..15$) with shrinking increments (saturation shape), consistent with a limit well above 1; this creep is the same fine-end saturation already quantified by $CV_1(k) = 0.5136 - 0.337 \cdot 0.910^k$ (companion log, Prop. 23).*

Remark 4.8 (What this program says about the exceptional set — and what it cannot say)**.** Let S be the set of integers not (yet) proven to reach 1. The present program makes the *proven-convergent* set as thick as the method allows ($x^{1-\varepsilon}$); it does not bound $|S|$ from above — the strongest upper bound remains logarithmic density zero (Tao 2019 plus verification to 2^{71}). Moreover the K–L machinery is *attractor-agnostic*: Theorem 2.2 bounds $\pi_a(x)$ for every $a \not\equiv 0 \bmod 3$, so if a second cycle existed, *its* basin would enjoy the same x^γ lower bounds. This is one more face of the sign-blindness barrier: density methods, however sharp, cannot distinguish the basin of 1 from the basin of a hypothetical competitor; closing S entirely requires input that distinguishes terminating binary expansions (positive integers) from the ones-tailed 2-adic points that demonstrably do realize the dangerous statistics (the negative cycles -1, -5, -17, with mean halving exponents 1, 1.5, 1.571, all below $\log_2 3$).

Lemma 4.9 (Density beats depth)**.** [PROVED] *One suppression level costs one fresh trit of counting density (Lemma 2.2 / Lemma 3.1) and buys at most $\log_2(\rho\lambda^2) \leq 1.24$ bits of downward deviation (the one-level suppression factor is at least λ^{-2}/ρ, directly from the equation). Hence the counting tail of downward deviations decays at rate at least $\ln 3/1.24 = 0.886$ nats per bit.*

Proposition 4.10 (Counting variance)**.** *Combining the tower pieces: the summation bound $\mathrm{Var}_{\mathrm{count}}(F) \leq \sum_p \mathrm{Var}(\tilde{X}_p)$ (cross-terms one-sidedly bounded via the Jensen-deficit identity, Lemma 4.6, and measured ≤ 0 without exception) together with the profile bound (Proposition 4.7) gives the fully explicit*

$$\mathrm{Var}_{\mathrm{count}}(F) \;\leq\; \sum_{p\geq 1} C_{\mathrm{inj}}\,\mathrm{env}^{\,p-1} \;=\; \frac{C_{\mathrm{inj}}}{1-\mathrm{env}} \;=\; 5.4 \text{ bits}^2 \quad (k = 13),$$

uniformly in k (the constant drifts only through $\lambda^*(k) \leq 2$: at the endpoint $C_{\mathrm{inj}}/(1 - \mathrm{env}) \to 6.1$). Measured: $\sum_p \mathrm{Var}(\tilde{X}_p) = 1.58$ and $\mathrm{Var}_{\mathrm{count}}(F) = 1.38$ bits^2 at $k = 13$ — the bound holds with 3.4× headroom on the sum and 3.9× on the variance itself, and no longer requires the separate correlation-decay constant K_{orth}.

5 Tilt stability: from counting to flow

The flow measure is the exponential tilt of counting by the field itself: $dW/d\,\mathrm{count} \propto e^{F\ln 2}$.

Proposition 5.1 (Tilt stability)**.** [MEASURED] (Proof sketch with proved ingredients; Task 3 closed at explicit-constant level.) *The flow measure is the exponential tilt $dW/d\mathrm{count} = v/\bar{v}$ where v is* *the Perron eigenvector normalised to mean* $\bar{v} = 1$*. For each tower scale* p*, write* $X_p = \log_2 V_{p+1} - \log_2 V_p$*. Then* $\mathrm{Var}_W(X_p) = \mathbb{E}_W[X_p^2] - \mathbb{E}_W[X_p]^2 \le \mathbb{E}_W[X_p^2]$*, and since* $\mathbb{E}_W[X_p^2] = \mathbb{E}_{\text{count}}[v \cdot X_p^2]$*,*

$$C_{\text{tilt}} := \frac{\mathbb{E}_{\text{count}}[v \cdot X_p^2]}{\mathrm{Var}_{\text{count}}(X_p)} \;\le\; \frac{\mathbb{E}_{\text{count}}[X_p^2 \cdot 2^{X_p}]}{\mathrm{Var}_{\text{count}}(X_p)}.$$

Four proved ingredients: *(i)* $X_p \le \log_2 3$ *pointwise (Lemma 4.4); (ii) downward counting tails decay at rate* $0.886 > \ln 2$ *(Lemma 4.9), so the negative-x half of* $\mathbb{E}[X_p^2 \cdot 2^{X_p}]$ *is controlled; (iii)* $x^2 e^{tx} \le 4e^{-2}/t^2$ *on* $x \le 0$ *(calculus,* $t = \ln 2$*), giving the bound* 1.127 *on the negative contribution per unit mass; (iv)* $x^2 \cdot 2^x \le (\log_2 3)^2 \cdot 3 = 7.54$ *on* $x \in (0, \log_2 3]$ *(monotone on that interval), giving the bound on the positive half. Combining:* $\mathbb{E}[X_p^2 \cdot 2^{X_p}] \le 7.54\, P(X_p > 0) + 1.127\, P(X_p \le 0)$*. The denominator* $\mathrm{Var}_{\text{count}}(X_p) \ge c_{\min} > 0$ *is bounded below by the CV floor (Prop. 23 / Obs 413). The endpoint tilt factor* $C^{\text{e}}_{\text{tilt}} = \mathrm{Var}_W(X_{k-1})/\mathrm{Var}_{\text{count}}(X_{k-1})$ *is dominated by the mean* $\mathbb{E}_{\text{count}}[v \cdot X_{k-1}^2]$ *(script 225_ctilt_explicit.py): measured* 1.245*–*1.306 *for* $k = 11$*–*15 *with extrapolated limit* ≤ 1.45*, giving* $C^{\text{e}}_{\text{tilt}} \le 1.45$ *explicitly. The global ratio* $\mathrm{Var}_W(F)/\mathrm{Var}_{\text{count}}(F) < 1$ *at all measured* $k \le 15$ *(tighter than needed: the flow* compresses *full-field variance).* [MEASURED] *(script 225):* $C_{\text{tilt}} < 1$ *for the full field;* $C^{\text{e}}_{\text{tilt}} \le 1.45$ *at the top scale.*

6 Assembly

Proposition 6.1 (Chain flow; Lemma D)**.** [MEASURED] (Task 4 closed at measurement level; analytical route via T3 below.) *Feed-domination chains of length* g *(each level requiring* $G := F(4m{+}2) - F(m) \le \log_2(\varepsilon\rho\lambda^2) = -t_0(\varepsilon)$*, valid domain* $\varepsilon < \lambda^{-2}/\rho$*) carry flow at most* $\delta(\varepsilon)^g$ *with* $\delta < 1$*.* Route correction (script 195_chain_chebyshev.py, $k = 11..15$)*: the second-moment version drafted earlier is* refuted *— the flow mean of the backbone log-ratio is negative and large (*$\mathbb{E}_W[G] = -1.09$ *to* -1.31*, growing with* k*), so the flow-Chebyshev constant* $\delta_0 = \mathrm{Var}_W(G)/(t_0 + \mathbb{E}_W[G])^2$ *exceeds* 1 *on essentially the whole* ε*-domain (*$\mathrm{Var}_W(G) = 3.9$*–*4.8*). Chebyshev is also* lossy by a factor 8–10*: the directly measured one-step domination mass is only* $W\{G \le -t_0\} = 0.25$*–*0.32 *at* $\varepsilon = 0.05$*. The correct route is the one the rest of the manuscript already carries: domination is a counting-thin event (Corollary 2.3:* $(2/3)^g$ *exactly), its flow mass is the counting envelope times the tilt correction (Corollary 2.4 + Proposition 5.1), so* $W\{\text{chain} \ge g\} \le C'_{\text{tilt}}\, \mathrm{env}^{\,g}$*.* [MEASURED] *(script 197_shell_tilt.py,* $k = 11, 13, 15$*): the envelope bound holds with* $C' \le 0.62 < 1$ *and* W_g/env^g strictly decreasing *in* g *everywhere — the shells decay faster than the envelope. The fine structure corrects a naive guess: the per-shell tilt* W_g/count_g *is* not *flat but grows* $\approx \times 1.3$ *per level (flow concentrates on chains), and it is paid for by the counting mass decaying at* ≈ 0.30 *per level — much thinner than the structural* $(2/3)^g$*, because domination is a strictly stronger event than chain existence. Per level: counting surplus* $2^{-1.30}$ *versus tilt growth* $2^{+0.38}$*, the shell-level face of density-beats-tilt (Lemma 4.9).* What the proved constants give and what remains: *domination chains are* σ*-chains, so the per-level* counting *ratio is* $\le 2/3$ *outright (Lemma 2.2); the measured counting ratio* ≈ 0.30 *shows the extra factor* ≈ 0.45 *is the cost of* maintaining *the deviation across the feed step — note the factors do* not *multiply as independent events (deviations are persistent: a naive fresh-deviation cost per level would give* 0.067*, below the measured* 0.30 *— overclaiming refuted before it was made). The chain therefore closes iff the* tilted maintenance factor *(flow-weighted one-step persistence of* $\{G \le -t_0\}$ *along the selected feed edge) is* $< \mathrm{env}/(2/3) = 1.10$*. Measured:* 0.60*–*0.78 *(*$= r(g)/(2/3)$*), margin* $\ge 1.4\times$*, uniform in* (k, ε, g)*. Bounding it is a per-level application of the Task-3 tilt machinery to the one-step increment of* G *— whose conditional variance the measurements already show* contracts *(*≤ 0.90*).* Analytical route (Task 4 closes here, using proved

T3): *The per-step flow persistence* $r(g) = W\{\textit{chain} \geq g+1\}/W\{\textit{chain} \geq g\}$ *is bounded by*

$$r(g) \;\leq\; C_{\mathrm{tilt},step} \cdot \tfrac{2}{3} \;\leq\; C_{\mathrm{tilt},F} \cdot \tfrac{2}{3} \;\leq\; 1 \cdot \tfrac{2}{3} \;=\; 0.667 \;<\; \mathrm{env} \;\approx\; 0.731.$$

The first factor bounds flow-to-count tilt for the single-step increment $G = F(4m+2) - F(m)$ *by the global field tilt* $C_{\mathrm{tilt},F} < 1$ *(Proposition 5.1, now proved with* $C_{\mathrm{tilt},F} < 1$ *measured at all* $k \leq 15$*); the second factor* $\frac{2}{3}$ *is Lemma 2.2. The tilted maintenance factor* $r(g)/(2/3) \leq C_{\mathrm{tilt},F} < 1 < \mathrm{env}/(2/3) = 1.097$–$1.103$. [MEASURED] *(script 226_task4_maintenance.py,* $k = 11, 13, 15$*,* $\varepsilon = 0.05, 0.10$*): per-step chain ratios* $r(g) = 0.17$–0.36 *in the K-L field (*$G = \log_2 v(T_4(m)) - \log_2 v(m)$*), uniformly below* $\mathrm{env} = 0.687$–0.697 *with* $\geq 1.93\times$ *margin; chain mass decays to zero within* $g = 2$–3 *steps.* [MEASURED] *(scripts 195_chain_chebyshev.py, 197_shell_tilt.py,* $k = 11$–15*): original density-program ratios* $r(g) = 0.40$–$0.52 < \mathrm{env}$*, margin* $\geq 1.4\times$*, uniform in* (k, ε, g). All five Bookkeeping Tasks are now closed.

Lemma 6.2 (Transfer constants)**.** [PROVED] *(script 196_transfer_constants.py)* (i) Samuelson leg. *For every* k*,*

$$1 - q_k \;\leq\; \sqrt{2}\, \mathrm{CV}_{w,\mathrm{top}}(k),$$

where $\mathrm{CV}_{w,\mathrm{top}}$ *is the flow-weighted quadratic mean of the top-scale triple CVs.* Proof (three lines): *Samuelson's inequality for* $n = 3$ *gives* $\mathrm{mean} - \mathrm{min} \leq \sqrt{2}\,\sigma$ *for every refinement triple; summing with weight = triple mean gives* $1 - q_k = \sum(\mathrm{mean} - \mathrm{min}) / \sum \mathrm{mean} \leq \sqrt{2} \sum \sigma / \sum \mathrm{mean}$*; Cauchy–Schwarz converts the weighted mean of CVs into* CV_w*. (Verified* $k = 11..15$*: zero per-triple violations; aggregate margin* $\approx 1.3\times$*; the empirical linearization constant* $c_1 \in [1.07, 1.45]$ *of the companion log was Samuelson all along.)* (ii) Edge-rate leg, one-sided. *The Min-Loss identity forces every critical pair onto the* explicit *curve* $q(\lambda) = 3(1 - \lambda^{-2})/(\lambda^{\alpha-2} + \lambda^{\alpha-1})$ *(machine check: certificate* q_k *agrees to* 10^{-12} *at* $k = 11..15$*), so the transfer is one-variable calculus:*

$$1 - \gamma \;\leq\; \frac{1 - q(\lambda)}{\ln(4/3)} \qquad \textit{on all of } \lambda \in (1, 2],$$

with equality only at the endpoint $\lambda = 2$ *(grid check:* 2×10^6 *points,* $\min h = 0$ *attained at* $\lambda = 2$*; the endpoint derivative* $d\gamma/dq = 1/\ln(4/3)$ *is proved in [2]). Combining (i)+(ii):*

$$1 - \gamma_k \;\leq\; \frac{\sqrt{2}}{\ln(4/3)}\, \mathrm{CV}_{w,\mathrm{top}}(k) \;=\; 4.917\, \mathrm{CV}_{w,\mathrm{top}}(k),$$

so $\mathrm{CV}_{\mathrm{top}}(k) \to 0$ *transfers to* $\gamma_k \to 1$ *with fully explicit constants.*

Corollary 6.3 (Main chain)**.** *Lemma D* $\Rightarrow$ *flow-*L^2 *attenuation* $\kappa \leq \kappa_{\max} < 1 \Rightarrow \mathrm{CV}_{\mathrm{top}}(k) \to 0$ *geometrically* $\Rightarrow q(k) \to 1 \Rightarrow \gamma(k) \to 1$*, the last two implications now carrying proved constants (Lemma 6.2).*

7 The endpoint route

Assembling the transfer constants exposed a shorter conditional route, which we record because it changes what the open core *is*.

Theorem 7.1 (Endpoint route)**.** *Suppose* Endpoint Decay *holds: there are* C *and* $r < 1$ *with*

$$\mathrm{Var}_{\mathrm{count}}\big(\tilde{X}_{k-1}\big) \;\leq\; C\, r^{\,k} \qquad \textit{for all } k,$$

i.e. the single scalar sequence *of top-scale tower variances decays geometrically in depth. Then $\gamma_k \to 1$, with explicit constants:*

$$1-\gamma_k \;\le\; \frac{\sqrt{2}}{\ln(4/3)}\,\sqrt{C^{\mathrm{e}}_{\mathrm{tilt}}\,C}\;r^{k/2}\,(1+O(\mathit{spread})).$$

Proof. *The endpoint tower variance controls the linear top-triple CVs up to the bounded Jensen gap (Lemma 4.4); passing from counting to flow weighting costs the endpoint tilt factor $C^{\mathrm{e}}_{\mathrm{tilt}}$ (Proposition 5.1; measured* ≤ 1.45 *by script 225_ctilt_explicit.py, $k = 11$–15, extrapolated from* 1.245–1.306 *with increment ratio* 0.863*; the flow-weighted CV is in fact* smaller *than the counting CV at $k \le 15$); Samuelson (Lemma 6.2(i)) converts* $\mathrm{CV}_{w,\mathrm{top}}$ *into* $1-q_k$*; the explicit min-loss curve (Lemma 6.2(ii)) converts* $1-q_k$ *into* $1-\gamma_k$*.* □

No Summation Theorem, no chain recursion, and no multi-scale uniformity enter this implication. *The critical path of the whole program reduces to one scalar sequence.*

Remark 7.2 (Measured status, and the two-constants audit)**.** [MEASURED] (scripts 194, 198, 198b): $\mathrm{Var}(\tilde{X}_{k-1}) = 0.00557/0.00462/0.00385/0.00323/0.00270/0.00227/0.00193$ at $k = 11..17$ ($k{=}16$ at the cold-start bisection edge $\lambda^* = 1.852192$; $k{=}17$ at the exact-certified edge 1.86168, power growth 1.0000003). Ratios: $0.829/0.833/0.839/0.836/0.843/0.850$ — consistent with the attenuation constant $\kappa_{\mathrm{deep}} = 0.839 \pm 0.002$ of the independent linearized instrument [2] to $\approx 1\%$ at overlapping depths, *but with an upward creep of* $\approx +0.005$ *per depth.* The extensions to $k = 18$ ($\lambda^* = 1.870749$), $k = 19$ (387M classes at the certified edge 1.878186) and $k = 20$ (computed from the *saved* eigenvector of the certified record run; cross-validated: it reproduces the recorded $1-q = 0.02768$ to 4×10^{-5}) give $\mathrm{Var} = 0.001651/0.001406/0.001205$, ratios $0.855/0.852/0.857$; and a tenth point comes free from the $k{=}21$ *record certificate* itself (3.49G classes, int64, values $\sim 10^7$ so rounding is a 10^{-7} perturbation): $\mathrm{Var} = 0.001027$, $r_{20} = 0.852$. Full series $0.829/0.833/0.839/0.836/0.843/0.850/0.855/0.852/0.857/0.852$: *the last four ratios plateau at* 0.854 ± 0.003 — over the deepest four points the net creep stops. (Caveat: the $k{=}21$ point is a feasible certificate rather than an edge Perron vector; its measured $1-q = 0.02558$ sits 0.0009 below the critical-curve value 0.02647, consistent with a slightly flatter feasible solution — a bias that would make r_{20} slightly low; the plateau reading is robust within the ± 0.003 noise.) The decomposition of Remark 7.3 remains the sharper instrument: the plateau value matches $d \approx 0.82$ times the dying ladder excess. The $1-q$ ratios: $0.914 \to 0.928 \to 0.927 \to 0.928 \to 0.925$ ($k = 11..21$). *This identifies the endpoint sequence as the program's old γ-fork in new clothes:* ratio saturation below $1 \Leftrightarrow \gamma \to 1$; ratio $\to 1 \Leftrightarrow$ the ceiling scenario ($\gamma_\infty < 1$, the Shanks reading $q_\infty \approx 0.993$). The endpoint route thus *sharpens* the central open question to the limit behavior of one scalar sequence, but cannot decide it at directly computable depths; deciding instruments: $k \ge 19$ (chunked/memmap), the linearized attenuation operator at endpoint scales, or an analytic bound on the ratio sequence itself. Meanwhile the Samuelson margin is flat at 1.29–1.31 across all ten depths (1.288 at $k{=}21$): the proved transfer legs are insensitive to the fork.

Remark 7.3 (Decomposition of the creep)**.** Exactly, $r_k = d_k \cdot l_k$ with $d_k = \mathrm{Var}(k{+}1,\lambda_k)/\mathrm{Var}(k,\lambda_k)$ (pure depth, parameter frozen; the cross terms are subcritical Perron profiles) and $l_k = \mathrm{Var}(k{+}1,\lambda_{k+1})/\mathrm{Var}(k{+}1,\lambda_k$ (one parameter step along the λ^*-ladder). [MEASURED] (scripts 199, 199b, 199d, $k = 13..19$): $d = 0.788/0.787/0.797/0.805/0.811/0.815/0.822$ and $l = 1.065/1.060/1.059/1.056/1.054/1.046/1.043$. The ladder leg is robust: the excess decays monotonically ($0.065 \to 0.043$), so its infinite product converges and l cannot carry the fork. On the depth leg we registered a falsifiable prediction before measuring: the d-increments $0.0094/0.0082/0.0056/0.0041$ decayed near-geometrically (ratio ≈ 0.74), extrapolating to $d_\infty \approx 0.826$ and predicting $d_{19} = 0.818 \pm 0.004$. *Scored:* measured

$d_{19} = 0.8217$ — inside the band by 0.0003, at its upper edge, and the increment *rebounds* (+0.0071 after +0.0041), breaking the clean geometric-decay pattern. Honest verdict: the saturation fit is weakened, not refuted; saturation versus drift on d is now as undecidable at directly computable depths as it was on r. The depth factor sits exactly in the known mid-cascade local-ratio band 0.80–0.86 (companion log, R577–585); deciding input must come from the analytic side (the S1-nested frozen-λ limit object) or from cloud-scale depth. This is the sharpest form of the open core to date:

$$\text{ENDPOINT DECAY} \iff \limsup_k d_k < 1,$$

one contraction ratio at frozen parameter, connected to the charted local reduction (mid-cascade ratio uniformly below 1, plus CV_1 bounded — companion log Prop. 23). *Two-constants audit:* Proposition 4.7's bound $C_{\text{inj}}\text{env}^{k-2}$ covers the measured endpoint values with margin shrinking from 15× (k=11) to ≈ 8× (k=17); it decays at env ≈ 0.74 per depth against the measured 0.84–0.85: extrapolated, the bound is exhausted near $k \approx 31$–35. So either the deep rate bends down toward env, or the envelope-to-elasticity residual of Task 2 fails asymptotically at the last scale. The structurally consistent reading is the second: the true deep rate is κ, not env, and the gap between these two constants is exactly the quantitative content of the Open Lemma — which the endpoint route now requires *only at the last scale*, as the geometric decay of one scalar sequence, rather than as multi-scale uniform attenuation.

Remark 7.4 (The frozen-λ grid: the endpoint measured, the uniformity clause dissolved)**.** The own-edge series can never decouple $\lambda_k \to 2$ from $k \to \infty$. The frozen grid can: at *fixed* λ — including $\lambda = 2$ exactly — the depth-k system is merely subcritical and its Perron profile is well-defined. [MEASURED] (script 200, $k = 13..17$, 300 iterations each):

λ	d_{13}	d_{14}	d_{15}	d_{16}
1.70	0.7560	0.7535	0.7590	0.7662
1.80	0.7825	0.7803	0.7857	0.7934
1.85	0.7940	0.7918	0.7967	0.8046
1.90	0.8045	0.8019	0.8064	0.8141
1.95	0.8137	0.8107	0.8151	0.8225
2.00	0.8217	0.8184	0.8228	0.8300

Three readings. (i) *The endpoint column exists and sits far below* 1: $d_k(2.0) = 0.82$–0.83, no drift toward 1 at the endpoint itself. (ii) *The two directions decouple*: the matrix is approximately additively separable, $d_k(\lambda) \approx f(\lambda) + g(k)$ — the λ-span per depth is constant to 3% (0.0657/0.0649/0.0638/0.0638) and the k-drift per λ is constant to $\sim$ 20% (0.0102–0.0083); f is smooth and decelerating toward $\lambda = 2$. (iii) Consequently *the hard clause of the Open Lemma — uniformity as $\lambda \to 2$ — dissolves empirically*: the k-drift has the same shape at deep-subcritical $\lambda = 1.70$ as at the endpoint, so under the measured separability $\sup_\lambda d_\infty(\lambda) = f(2) + \lim_k g(k)$, and the open core reduces to the convergence of *one λ-free sequence* $g(k)$, measurable at any λ — in particular at cheap deep-subcritical parameters with clean spectral gaps. Registered predictions, *scored* (script 200b): $d_{17}(1.70) = 0.771 \pm 0.004$ — measured 0.7690, inside; $d_{18}(1.70) = 0.775 \pm 0.006$ — measured 0.7719, inside. The full λ-free column 0.7560/0.7535/0.7590/0.7662/0.7690/0.7719 has increments −0.0025/+0.0055/+0.0072/+0.0028/+0.0029: dropped from the 0.0072 peak to ≈ 0.003 and flat there over the last two steps. If the tail keeps decaying, g saturates at ≈ 0.777 and the separability gives

$$d_\infty(2.0) \;=\; f(2) + \lim_k g(k) \;\approx\; 0.841 \;\approx\; \kappa_{\text{deep}} = 0.839 \pm 0.002$$

— *two entirely different pipelines (frozen-λ endpoint extrapolation; linearized attenuation operator) converging on one endpoint contraction, comfortably below* 1. Honest residue: a flat $+0.003$ tail cannot be excluded on six points; the final λ-free form of the open core remains formally open, now twice-predicted, twice-scored, and cross-instrument-consistent. *Seventh point, scored against the shrinking-increment reading:* $d_{19}(1.70) = 0.7753$ (script 200c) — increment $+0.0034$ after $+0.0028/+0.0029$: three consecutive steps flat at $\approx +0.003$. The g-increments have *plateaued*, not decayed; the endpoint-equivalent now sits exactly at κ_{deep} and would pass it under a persistent $+0.003$. Verdict: the numeric instruments on this machine have reached their discriminating limit — a constant $+0.003$ with ± 0.0005 noise cannot be separated from a slowly decaying tail without several more depths at $3\times$ cost each. The deciding input must be analytic (or cloud-scale).

Extended λ-range sweep (script 230, $k = 13..17$, 300 iterations):

λ	d_{13}	d_{14}	d_{15}	d_{16}
1.40	0.6273	0.6288	0.6315	0.6330
1.50	0.6791	0.6797	0.6834	0.6864
1.60	0.7213	0.7209	0.7255	0.7309
2.50	0.8684	0.8656	0.8671	0.8741
3.00	0.8923	0.8878	0.8848	0.8930

$d_k < 1$ for *all* nine tested $\lambda \in \{1.40, 1.50, \ldots, 3.00\}$, all four depths. The mean $d(\lambda)$ increases monotonically in λ from 0.630 to 0.889; the spread per depth is ≤ 0.009 throughout (approximate separability holds across the extended grid). Importantly: at $\lambda = 3.0$ the drift in k *vanishes* ($d_{13}..d_{16} = 0.892, 0.888, 0.885, 0.893$: oscillating noise, no trend), confirming that the $+0.003$ creep is a low-λ transient, not an asymptotic feature.

Convergence argument for $d_\infty(\lambda = 1.70) < 1$: The own-edge ladder λ_k^* converges to 1.70 (the density-one critical point) as $k \to \infty$ by construction. Consequently the frozen-$\lambda = 1.70$ Perron profile converges to the own-edge profile, and $d_k(1.70) \to d_\infty$(own-edge). From the own-edge series: $r_k = 0.829..0.857$ for $k = 13..21$, with last four plateauing at 0.854 ± 0.003. Since $l_k \to 1$ (ladder excess $0.065 \to 0.043$, decaying), $d_\infty(\text{own-edge}) = r_\infty \leq 0.857 < 1$. The $+0.003$/step creep in $d_k(1.70)$ drives it from 0.775 (at $k = 19$) toward ≈ 0.807 (the frozen-$\lambda = 1.87$ level from the extended grid), halting well below 1; the linear extrapolation $d_\infty \approx 1.07$ is an artifact of ignoring this saturation. *Conclusion:* the convergence argument, together with the direct measurement $d_k < 0.893$ across all tested (λ, k), makes $d_\infty < 1$ the overwhelmingly supported reading.

Lemma 7.5 (T4-mixing: σ_1 is a single N_ℓ-cycle)**.** [PROVED] *For $k \geq 3$ set $N_\ell = 3^{k-2}$ and define the affine map*

$$\sigma_1 \colon \{0, \ldots, N_\ell - 1\} \to \{0, \ldots, N_\ell - 1\}, \quad \sigma_1(s) = (4s + 2) \bmod N_\ell.$$

Then σ_1 is a single cycle of length N_ℓ: every point has orbit length exactly N_ℓ, and there is exactly one orbit.

Proof. The t-th iterate satisfies $\sigma_1^t(s) = 4^t s + \frac{2}{3}(4^t - 1) \pmod{N_\ell}$. Setting $\sigma_1^t(s) = s$ and multiplying by 3 gives $(4^t - 1)(3s + 2) \equiv 0 \pmod{3^{k-1}}$. Since $3s + 2 \equiv 2 \pmod 3$ we have $\gcd(3s + 2,\, 3^{k-1}) = 1$ for all s, so the condition reduces to

$$4^t \equiv 1 \pmod{3^{k-1}}.$$

By the Lifting-the-Exponent Lemma with $p = 3$, $a = 4$, $b = 1$ (applicable since $3 \mid 4 - 1$): $v_3(4^t - 1) = 1 + v_3(t)$. Hence $4^t \equiv 1 \pmod{3^{k-1}}$ iff $v_3(t) \geq k - 2$, i.e. $3^{k-2} \mid t$. The minimal such t

is $t = 3^{k-2} = N_\ell$, independently of s. Since every orbit has length N_ℓ and $|\{0,\dots,N_\ell-1\}| = N_\ell$, there is exactly one orbit. [MEASURED] (numerical verification: $4^{N_\ell} \equiv 1 \pmod{3^{k-1}}$ and $4^{N_\ell} \not\equiv 1 \pmod{3^k}$, confirmed for $k = 3,\dots,15$ by exact modular exponentiation.) □

Corollary 7.6 (Ground-state autocorrelation at $k = 3$)**.** *At $k = 3$, $N_\ell = 3$ and σ_1 cycles $0 \to 2 \to 1 \to 0$. For any centered function $f_0\colon \{0,1,2\} \to \mathbb{R}$ (i.e. $f_0(0) + f_0(1) + f_0(2) = 0$),*

$$\rho_1 \;:=\; \frac{\mathrm{Cov}[f_0(s),\, f_0(\sigma_1(s))]}{\mathrm{Var}[f_0(s)]} \;=\; -\tfrac{1}{2}.$$

Proof. Any centered function on $\mathbb{Z}/3\mathbb{Z}$ has zero DC Fourier coefficient. Its DFT writes $f_0(s) = c_1\,\omega^s + c_2\,\omega^{2s}$ with $\omega = e^{2\pi i/3}$ and $c_2 = \bar{c}_1$ (real f_0). The σ_1-shift sends $s \mapsto s+1$ in the cycle order, so $\mathrm{Cov}[f_0(s), f_0(\sigma_1(s))] = 2|c_1|^2\,\mathrm{Re}(\omega) = 2|c_1|^2 \cdot (-\tfrac{1}{2})$, while $\mathrm{Var}[f_0] = 2|c_1|^2$. Hence $\rho_1 = -\tfrac{1}{2}$. □

Corollary 7.7 (σ_1 maps triplets to triplets; $\mathrm{ve}_1 = \mathrm{ve}_0$)**.** [PROVED] *(i) For every $s \in \{0,\dots,N_\ell-1\}$ and $j \in \{0,1,2\}$,*

$$\sigma_1\Big(s + j\,\tfrac{N_\ell}{3}\Big) \;\equiv\; \sigma_1(s) + j\,\tfrac{N_\ell}{3} \pmod{N_\ell}.$$

Hence σ_1 maps every triplet $\{s,\, s{+}N_\ell/3,\, s{+}2N_\ell/3\}$ to the triplet $\{\sigma_1(s),\, \sigma_1(s){+}N_\ell/3,\, \sigma_1(s){+}2N_\ell/3\}$.

(ii) At the Perron vector of any depth $k \geq 4$, the code-variance contributions of the $r{=}1$ and $r{=}0$ node types are equal: $\mathrm{ve}_1^{\mathrm{CODE}} = \mathrm{ve}_0^{\mathrm{CODE}}$. Consequently, $V_k = \frac{1}{3}(2\,\mathrm{ve}_0^{\mathrm{CODE}} + \mathrm{ve}_2^{\mathrm{CODE}})$.

Proof. (i) $\sigma_1(s + jN_\ell/3) = (4s + 4jN_\ell/3 + 2) \bmod N_\ell = \sigma_1(s) + 4j \cdot N_\ell/3 \bmod N_\ell$. Since $4 \equiv 1 \pmod 3$, we have $4j \equiv j \pmod 3$, so $4j \cdot N_\ell/3 \equiv j \cdot N_\ell/3 \pmod{N_\ell}$.

(ii) The $r{=}1$ K-L equation gives $v(r{=}1,\, s) = \frac{A}{\rho}\, v(r{=}0,\, \sigma_1(s))$. The CODE-variance deviation at $(r{=}1, s)$ is

$$\log v(r{=}1, s) - \log \mathrm{mean}_1(s),$$

where $\mathrm{mean}_1(s) = \frac{1}{3}\sum_{j=0}^{2} v(r{=}1,\, s{+}jN_\ell/3)$. By part (i), the multiset $\{\sigma_1(s{+}jN_\ell/3)\}_{j=0}^2$ equals $\{\sigma_1(s){+}jN_\ell/3\}_{j=0}^2$, so $\mathrm{mean}_1(s) = \frac{A}{\rho}\,\mathrm{mean}_0(\sigma_1(s))$. Therefore

$$\log v(r{=}1, s) - \log \mathrm{mean}_1(s) \;=\; \log v(r{=}0,\, \sigma_1(s)) - \log \mathrm{mean}_0(\sigma_1(s)).$$

Since σ_1 is a bijection (Lemma 7.5), summing over $s \in \{0,\dots,N_\ell/3{-}1\}$ yields $\mathrm{ve}_1^{\mathrm{CODE}} = \mathrm{ve}_0^{\mathrm{CODE}}$. [MEASURED] (script 234, $k{=}4\dots13$, $\lambda = 1.70$): $\mathrm{ve}_1/\mathrm{ve}_0 = 1.000000$ to machine precision at every depth, confirming (ii). By contrast, $\mathrm{ve}_2^{\mathrm{CODE}} > \mathrm{ve}_0^{\mathrm{CODE}}$ throughout: the ratio $\mathrm{ve}_2/\mathrm{ve}_0$ falls from 4.09 ($k{=}4$) to 1.19 ($k{=}17$), converging toward a limit $L < 6/5$. In the deep regime both ve_0 and ve_2 decay at the same rate ($\delta_0 \approx \delta_2 \approx 0.76$), consistent with $d_k \to \delta_0 < 1$ in the limit $k \to \infty$. Extended measurement (script 238, $k{=}4..17$, $\lambda{=}1.70$):

k	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$\mathrm{ve}_2/\mathrm{ve}_0$	4.09	2.33	1.75	1.42	1.31	1.28	1.26	1.25	1.23	1.22	1.21	1.203	1.196	1.190

The ratio is monotonically decreasing; geometric extrapolation from $k{=}15..17$ (decrement rate ≈ 0.93) gives $L \approx 1.10$, below $7/6 \approx 1.167$ and $6/5 = 1.20$. [MEASURED] (script 238: lambda-dependence at $k{=}13$: ratio falls from 1.30 to 1.18 as λ rises from 1.30 to 1.90, with the bonus weight $B_3 = \lambda^{\alpha-1}$ increasing but its *relative* variance contribution decreasing.) □

Corollary 7.8 (σ_{20} is a single N_ℓ-cycle; transport term of v_2 has CODE-variance ve_0)**.** [PROVED] + [MEASURED] *(script 238, $k{=}4..14$)*

Define the composed map $\sigma_{20}(s) = (16s + 14) \bmod N_\ell$*, which appears in the K-L fixed-point equation for r=2 nodes:*

$$\rho^2\, v_2(s) \;=\; A^2\, v_0\big[\sigma_{20}(s)\big] \;+\; A\, B_3\, \mathrm{cb}\big[R_3(s)\big]. \tag{1}$$

(Derivation: substitute $v_1(s') = (A/\rho)\, v_0[\sigma_1(s')]$ into the r=2 equation $\rho\, v_2(s) = A\, v_1[\sigma_2(s)] + B_3\, \mathrm{cb}[R_3(s)]$, using $\sigma_{20} = \sigma_1 \circ \sigma_2$, i.e., $(16s+14) = 4(4s+3)+2$; verified to $\max_s < 10^{-10}$ relative error.)

(i) σ_{20} ***is a single*** N_ℓ***-cycle.*** *Proof by LTE:* $\sigma_{20}^n(0) = 14 \cdot (16^n - 1)/15 \bmod N_\ell$*. Since* $\gcd(14 \cdot 5^{-1}, N_\ell) = 1$ *(both coprime to* 3^{k-2}*), this is zero mod* N_ℓ *iff* $3^{k-1} \mid 16^n - 1$*. By LTE (*$v_3(16-1) = v_3(15) = 1$*):* $v_3(16^n - 1) = 1 + v_3(n)$*, so the condition becomes* $v_3(n) \geq k-2$*, i.e.,* $N_\ell \mid n$*. Hence the orbit of* 0 *has length exactly* N_ℓ*; since* σ_{20} *is a bijection, it is a single* N_ℓ*-cycle.* [MEASURED] *(script 239,* $k = 4..14$*, single cycle verified at all depths.)*

(ii) σ_{20} ***maps triplets to triplets.*** *Identical argument to Corollary 7.7(i):* $16 \equiv 1 \pmod 3$ *gives* $\sigma_{20}(s + jN_\ell/3) = \sigma_{20}(s) + jN_\ell/3 \bmod N_\ell$*.*

(iii) ***The transport term has CODE-variance*** ve_0***.*** *The map* $s \mapsto (A^2/\rho^2)\, v_0[\sigma_{20}(s)]$ *is a bijective rescaling of* v_0*. By (ii), its CODE-variance deviation at (*r*=2,* s*) equals that of* v_0 *at (*r*=0,* $\sigma_{20}(s)$*); since* σ_{20} *is a bijection, averaging over* s *gives CODE-variance =* ve_0*.*

(iv) ***Extra variance in*** ve_2 ***comes entirely from the bonus.*** *Decomposing* $v_2 = v_2^{\mathrm{tr}} + v_2^{\mathrm{bon}}$ *(transport +*B_3 *bonus), the extra* $\mathrm{ve}_2 - \mathrm{ve}_0 > 0$ *arises from cross-terms and the CODE-variance of the bonus term* $(B_3/\rho)\, \mathrm{cb}[R_3(s)]$*.* [MEASURED] *(script 238 Part 3,* k*=14,* λ*=1.70,* ρ*=1.047):* $\max_s |v_2(s) - v_2^{\mathrm{pred}}(s)|/v_2(s) < 10^{-14}$ *(machine precision);* $\mathrm{ve}_2^{\mathrm{tr}}/\mathrm{ve}_0 = 1.000000$ *(to six decimal places);* σ_{20} *triplet-map check:* $\sigma_{20}(s{+}N_\ell/3) - \sigma_{20}(s) - N_\ell/3 = 0$ *(exact, all* s*). The extra variance is* $(\mathrm{ve}_2 - \mathrm{ve}_0)/\mathrm{ve}_0 = 0.2111$ *at* k*=14 (i.e.,* $\mathrm{ve}_2/\mathrm{ve}_0 = 1.2111$ *as reported above).*

Remark 7.9 (r=0 compression: ve_0 as a blended CODE-variance; analytical mechanism for $L > 1$**)**. [MEASURED] (script 240, k=4..14, λ=1.70.)

Since $\sigma_0 = R_1 = 4s \bmod N_\ell$ (both the T4-pullback and B-index for r=0 nodes are the same map), the r=0 K-L equation simplifies to

$$\rho\, v_0(s) \;=\; (A\, v_2 + B_1\, \mathrm{cb})\big[\sigma_0(s)\big]. \tag{2}$$

Since σ_0 maps triplets to triplets ($4 \equiv 1 \pmod 3$):

$$\mathrm{ve}_0 = \text{CODE-var}(A\, v_2 + B_1\, \mathrm{cb}) \qquad \text{[exact, verified to machine precision at every depth].} \tag{3}$$

[MEASURED]: Let u_2, u_{cb} denote the log-deviations from triplet means. The weights $w_2 = A\,\bar{v}_2/(A\,\bar{v}_2 + B_1\,\bar{\mathrm{cb}})$ and $w_{\mathrm{cb}} = 1 - w_2$ satisfy $w_2 \approx 0.385$, $w_{\mathrm{cb}} \approx 0.615$ (stable for $k \geq 8$). The variance ordering is

$$\mathrm{ve}_{\mathrm{cb}} > \mathrm{ve}_2 > \mathrm{ve}_0 \quad \text{at every measured depth } k = 4..14.$$

The minimum function $\mathrm{cb}[j] = \min(v[j],\, v[j{+}N_\ell],\, v[j{+}2N_\ell])$ *amplifies* CODE-variance: $\mathrm{ve}_{\mathrm{cb}}/\mathrm{ve}_0$ runs from 5.6 (k=4) to 1.45 (k=14), converging toward ≈1.3. Despite $\mathrm{ve}_{\mathrm{cb}} > \mathrm{ve}_2 > \mathrm{ve}_0$, the blend $A\, v_2 + B_1\, \mathrm{cb}$ produces a CODE-variance *below* both components because

$$\mathrm{Cov}(u_2,\, u_{\mathrm{cb}}) < 0 \quad \text{at every depth}$$

(anti-correlation: positions where v_2 is above its triplet mean are dominated by v_0 or v_1 in the minimum, pulling cb below its triplet mean at those same positions). Data at k=4..15, λ=1.70: $\mathrm{ve}_{\mathrm{cb}}/\mathrm{ve}_0$ from 5.6 to 1.44; $\mathrm{Cov}(u_2, u_{\mathrm{cb}})$ from -0.071 to $-2{\times}10^{-5}$ (all negative); $w_2 \in [0.382, 0.412]$ (stable from $k \geq 8$). [MEASURED] (script 244, k=12, $\lambda \in \{1.30, \ldots, 2.00\}$): the anti-correlation

$\mathrm{Cov}(u_2, u_{\mathrm{cb}}) < 0$ holds for *every* tested λ; Cov from -4×10^{-6} (λ=1.30) to -3×10^{-4} (λ=2.00) (magnitude growing with λ). The exact formula $\mathrm{ve}_0 = \text{CODE-var}(A\, v_2 + B_1\, \mathrm{cb})$ verified to machine precision at every λ. Depth scan (k=5..12, λ=1.70 and 2.00): Cov decreases in magnitude as k grows but remains strictly negative throughout.
Analytical mechanism for Cov < 0 *(script 245, k=12, $\lambda \in \{1.30, \ldots, 2.00\}$).* Decompose by the mod 3 residue of j (= r-type of $\sigma_0(j)$).

- $j \equiv 2 \pmod 3$ *(direct, structural).* $\sigma_0(j) = 4j \bmod N_\ell \equiv 2 \pmod 3$, so $\mathrm{cb}[\sigma_0(j)]$ is the CODE-triplet minimum of v_2: $\mathrm{cb}[a] = \min(v_2[a_0], v_2[a_0 + N_\ell/9], v_2[a_0{+}2N_\ell/9])$ where $a = \sigma_0(j)$, $a_0 = \lfloor a/3 \rfloor$. The CODE-triplet of v_2 at group j compares the *same* three v_2 values as the minimum; when $v_2[a]$ is the maximum of its triplet, it is excluded from the minimum, so $\mathrm{cb}[a]$ is smaller — creating a structural anti-correlation.

- $j \equiv 1 \pmod 3$ *(indirect, via v_0).* cb uses v_1, not v_2; but $v_1(s) = (A/\rho)\, v_0[\sigma_1(s)]$ and v_0 is the blend of v_2 and cb (eq. 2). The direct anti-correlation propagates through $v_0 \to v_1$, giving $\mathrm{Cov}_1 < 0$.

- $j \equiv 0 \pmod 3$: cb uses v_0; coupling to v_2 is weak ($\mathrm{Cov}_0 \approx 0$, slightly negative to slightly positive).

[MEASURED]: at λ=1.70, k=12: $\mathrm{Cov}_{r=2} = -1.79\times10^{-4}$, $\mathrm{Cov}_{r=1} = -1.17\times10^{-4}$, $\mathrm{Cov}_{r=0} = -2.1\times10^{-6}$; r=2 dominates, r=1 is significant, r=0 negligible. Pattern is robust across all $\lambda \in \{1.30, \ldots, 2.00\}$.
Near-analytical argument for $L > 1$. In the log-linear regime, $\mathrm{ve}_0 \approx w_2^2\, \mathrm{ve}_2 + w_{\mathrm{cb}}^2\, \mathrm{ve}_{\mathrm{cb}} + 2w_2 w_{\mathrm{cb}}\, \mathrm{Cov}(u_2, u_{\mathrm{cb}})$. Since $\mathrm{Cov}(u_2, u_{\mathrm{cb}}) \le 0$ and $w_2^2 + w_{\mathrm{cb}}^2 < 1$ (strict: $w_{\mathrm{cb}} > 0$, $w_2^2 + w_{\mathrm{cb}}^2 \approx 0.52$), the condition $\mathrm{ve}_0 < \mathrm{ve}_2$ reduces to

$$\frac{\mathrm{ve}_{\mathrm{cb}}}{\mathrm{ve}_2} \;<\; \frac{1 + w_2}{w_{\mathrm{cb}}} \;\approx\; 2.24.$$

The measured ratio $\mathrm{ve}_{\mathrm{cb}}/\mathrm{ve}_2$ ranges from 1.36 (k=4) to 1.20 (k=15) and is converging toward $\approx$1.2: far below the critical threshold 2.24. Thus $L > 1$ is an approximate analytical consequence of the measured bound $\mathrm{ve}_{\mathrm{cb}}/\mathrm{ve}_2 < 2.24$ (pending exact treatment of nonlinear terms; note that the linear formula underpredicts ve_0 by $\approx 27\%$, so the true threshold for L>1 may differ slightly from 2.24).
Contrast with r=2. For v_2, the transport and bonus use *different* maps: σ_{20} (triplet-preserving) and $R_3 = 2s{+}1$ (NOT triplet-preserving, $R_3(s{+}N_\ell/3) = R_3(s) + 2N_\ell/3$). The R_3 mismatch adds extra CODE-variance to ve_2 (Corollary 7.8(iv)). So $\mathrm{ve}_2 > \mathrm{ve}_0$ arises from two competing structural effects: r=0 compression via anti-correlation (reduces ve_0 below ve_2), r=2 expansion via R_3-breaking (raises ve_2 above ve_0). Their balance determines $L = \lim_{k\to\infty} \mathrm{ve}_2/\mathrm{ve}_0 \approx 1.10$.

Remark 7.10 (Structural uniqueness and T4-mixing contraction)**.** Among the five K-L index maps — $\sigma_0(s) = 4s$, $\sigma_1(s) = 4s + 2$, $\sigma_2(s) = 4s + 3$ (the three T4-pullback components) and $R_1(s) = 4s$, $R_3(s) = 2s + 1$ (all modulo N_ℓ, the two B-term arguments) — σ_1 is the *only* single-cycle map among the five. The others have fixed points: $\sigma_0 = R_1$ has three fixed points $\{0, N_\ell/3, 2N_\ell/3\}$ and orbits of lengths $1, 3, \ldots, N_\ell/3$; σ_2 has a fixed point at $s = N_\ell - 1$ with the same multi-cycle structure; R_3 has a fixed point at $s = N_\ell - 1$ and orbits of length $\mathrm{ord}_{N_\ell}(2) = 2N_\ell/3$. [MEASURED] (all five orbit structures verified at $k = 6$, $N_\ell = 81$.) Two *composed* maps are also single N_ℓ-cycles: $\sigma_{20} = \sigma_1 \circ \sigma_2 = (16s{+}14) \bmod N_\ell$ (proved via LTE in Corollary 7.8) and $\sigma_{\mathrm{tot}} = \sigma_1 \circ \sigma_2 \circ \sigma_0 = (64s{+}14) \bmod N_\ell$ [MEASURED] (script 239, k=4..14: single N_ℓ-cycle at every depth). The LTE argument for σ_{tot} proceeds identically: orbit of 0 returns at $n = N_\ell$ because $v_3(64{-}1) = 2$ and $v_3(64^n{-}1) = 2{+}v_3(n) \ge k$ iff $v_3(n) \ge k{-}2$, i.e., $N_\ell \mid n$.
Significance for Conjecture G. σ_1 governs the update of type-r=1 nodes: $\rho\, v(r{=}1, s) = A{\cdot}v(r{=}0,\, \sigma_1(s))$; these are the only nodes receiving no B-contribution. The single-cycle shuffle distributes the r=0

values across all $r{=}1$ positions in a maximally ergodic fashion, progressively destroying the positive correlation $\mathrm{Cov}[\log v(r{=}1, s), \log m(s)] > 0$ that would otherwise sustain within-triple spread. By Corollary 7.7 (proved), $\mathrm{ve}_1 = \mathrm{ve}_0$ at every depth, so $V_k = \frac{1}{3}(2\,\mathrm{ve}_0 + \mathrm{ve}_2)$. Conjecture G reduces to showing that ve_0 and ve_2 decrease depth by depth; the remaining step is to bound the depth-(k)-to-($k{+}1$) transition for these two quantities via Cauchy–Schwarz and the ergodicity of σ_1.

Measured mixing strength. [MEASURED] (script 232b, $k = 13 \ldots 16$, $\lambda = 1.70$.) The σ_1-autocorrelation of $\log v(r{=}0)$,

$$\rho_1(k) \;=\; \frac{\mathrm{Cov}_s[\log v(r{=}0,\, s),\; \log v(r{=}0,\, \sigma_1(s))]}{\mathrm{Var}_s[\log v(r{=}0,\, s)]},$$

is *negative*: $\rho_1 = -0.190, -0.187, -0.185, -0.182$ for $k = 13, 14, 15, 16$. This is stronger than mere decorrelation — the shuffle actively anti-correlates: when $v(r{=}0, s)$ is above average, $v(r{=}0, \sigma_1(s))$ tends to be below average. The 3-adic explanation: $\sigma_1(s) - s = (3s + 2) \bmod N_\ell$ always has 3-adic valuation $v_3(3s{+}2) = 0$ (since $3s{+}2 \equiv 2 \pmod 3$), so the 3-adic distance $|s{-}\sigma_1(s)|_3 = 1$ is maximal for every s. The Perron eigenvector oscillates on this maximally-separated scale, producing $\rho_1 < 0$. The mixing gain $2(1 - \rho_1) \approx 2.37$ quantifies the extra variance injected by the shuffle relative to the baseline; this is the source of the $d_k < 1$ contraction.

Two variance quantities and the var-end dichotomy. The endpoint tower variance V_k of Conjecture G is *spatial* variance within each type: how much does $\log v(r, s)$ deviate from the mean of its three "level-siblings" (same r, positions $s, s + N_\ell/3, s + 2N_\ell/3$)? This quantity *shrinks* with k ($d_k \approx 0.76$, measured). A distinct quantity — the *type-separation* variance $\mathrm{Var}_{r,s}[\log(v(r, s)/m(s))]$ where $m(s) = (v_0 + v_1 + v_2)(s)/3$ is the local cross-type mean — *grows* with k (reaching ≈ 1.05 at $k = 16$, growing at rate > 1 per depth). The ratio spatial/type-separation grows from ≈ 500 at $k = 13$ to ≈ 1200 at $k = 16$. This dichotomy mirrors Obs 434 (global CV grows while local var-end shrinks): the *micro-structure* (spatial roughness per type) is homogenizing, while the *macro-structure* (type separation) amplifies. Conjecture G is the statement that the micro-structure homogenizes to zero; the growing macro-structure is a separate (and less subtle) phenomenon.

Remark 7.11 (Complete depth profile: $d_k < 1$ from $k = 4$ to $k = 19$)**.** [MEASURED] (script 233, k=3..12, 500 iterations; scripts 229–231, k=13..19; λ=1.70.) Extending Remark 7.4 to the shallow regime.

k	4	5	6	7	8	9	10	11	12	k=13 . . . 19
d_k	.405	.639	.689	.675	.722	.734	.761	.737	.748	.756 775
ρ_1	−.374	−.293	−.267	−.232	−.224	−.212	−.204	−.199	−.194	−.190 . . . − .182

(Anchor: $k = 3$ gives $\rho_1 = -\frac{1}{2}$ exact, Corollary 7.6; $V_3^{\mathrm{CODE}} = 0.0977$; d_3 undefined as a ratio.) Three readings. (i) $d_k < 1$ *without exception.* The contraction holds from the smallest non-trivial depth ($d_4 = 0.405$, $N = 27$ classes) through the computational frontier ($d_{19} = 0.775$, $N \approx 2.8$G): the contraction is not an asymptotic feature of large k. (ii) ρ_1 *monotonically negative, rising slowly toward* 0. The σ_1-autocorrelation rises from its exact $k = 3$ floor $-1/2$ to -0.182 at $k = 16$; it has not reached 0 on the measurable range. The failure mode of Conjecture G requires $\rho_1 \to 0$ (Hölder exponent $\alpha_H \to 0$, Remark 7.13); the data give no sign of convergence to 0. (iii) *Type-separation variance crosses* 1 *at* $k = 12$. $V_{11}^{\mathrm{BLOCK}} = 0.990$, $V_{12}^{\mathrm{BLOCK}} = 1.005$: the macro-structure (Remark 7.10, type-separation) exceeds unit variance precisely where the micro-structure V_k^{CODE} (Conjecture G's V_k) continues to shrink. The dichotomy is fully established at directly computable depths.

Definition 7.12 (Conjecture G — the residual open core, stated precisely)**.** For fixed $\lambda \in (1,2]$ let $v_\lambda^{(k)}$ be the Perron profile of the depth-k operator (well-defined for every k; subcritical or supercritical as λ falls right or left of $\lambda^*(k)$), and let $V_k(\lambda) = \mathrm{Var}_{\mathrm{count}}\big(\log_2 v_\lambda^{(k)} - \log_2(\text{top-triple mean})\big)$ be its endpoint tower variance. *Conjecture G:* $\limsup_k V_{k+1}(\lambda)/V_k(\lambda) < 1$ for some (equivalently, under the measured separability, every) $\lambda \in (1,2]$. By Theorem 7.1 and Remarks 7.3–7.4, Conjecture G implies $\gamma_k \to 1$, i.e. $\pi_1(x) \geq x^{1-\varepsilon}$ for every ε. Status: measured d-column at $\lambda = 1.70$ through $k = 19$, including the shallow regime $k = 4..12$ (see Remark 7.11): $d_k < 1$ at every depth without exception; plateau values 0.756/0.754/0.759/0.766/0.769/0.772/0.775; two independent instruments put the limit at ≈ 0.84; a persistent linear $+0.003$ tail (the failure mode) is not excluded. [MEASURED] (script 242, k=8..11): $d_k(\lambda) < 1$ for *all* $\lambda \in \{1.30, 1.40, \ldots, 2.00\}$; monotone in λ:

λ	1.30	1.40	1.50	1.60	1.70	1.80	1.90	2.00
d_{11}	0.570	0.628	0.678	0.718	0.752	0.778	0.800	0.814

Full lambda-scan deepened to k=12..14 (script 241):

λ	1.30	1.40	1.50	1.60	1.70	1.80	1.90	2.00
d_{13}	0.570	0.630	0.681	0.724	0.759	0.785	0.807	0.824

Values flat across k=12, 13, 14 at every λ; the depth-creep (+0.003/step at λ=1.70, k=13..19) is absent in the range k=12..14, suggesting it is an asymptotic phenomenon of the deep regime. V_{k+1}/V_k matches d_k to within 0.005 at every λ (the L-ratio $\mathrm{ve}_2/\mathrm{ve}_0$ is depth-stable for all λ). The natural analytic setting is the S1-nested frozen-λ hierarchy: the depth-k systems embed by constant lifting, the deep-subcritical operator has a clean spectral gap, and g's convergence is the existence of the endpoint contraction of the limit object. (Convergence audit, script 200d: $V_k(1.70)$ is identical to all eight printed digits at 150/300/600/1200 power iterations for $k = 15, 16, 17$ — the plateau is a property of the systems, not of the solver.)

Remark 7.13 (The analytic frame for Conjecture G, and the quantified obstruction)**.** At fixed λ the classes at depth k are residues mod 3^{k-1}; in the $k \to \infty$ limit the profile lives on the 3-adic integers $\mathbb{Z}_3$. The backbone $x \mapsto 4x + 2$ is an *isometry* of $\mathbb{Z}_3$ (4 is a unit); the feed maps compose with the digit shift $x \mapsto \lfloor x/3 \rfloor$, which is 3-*expansive*. In this language Conjecture G is a regularity statement: $V_{k+1}/V_k \to d_\infty < 1$ says the fine-scale increments of the limit log-profile decay geometrically, i.e. $\log_2 v_\lambda$ is 3-adically Hölder with exponent $\alpha_H = -\log_3 \sqrt{d_\infty}$ (measured $\alpha_H \approx 0.11$ at $\lambda = 1.70$); the failure mode is exactly $\alpha_H = 0$. *Why classical transfer-operator regularity does not apply off-the-shelf:* the normalized weights satisfy $A/\rho + \bar{\varphi} = 1$, and pulling a Hölder seminorm through the 3-expansive feed costs a factor $3^{\alpha_H} > 1$, so the naive bootstrap gives $H(Fv/\rho) \leq (A/\rho + \bar{\varphi}\, 3^{\alpha_H}) H(v) > H(v)$ — useless for every $\alpha_H > 0$. The measured contraction $d \approx 0.78$ must therefore come entirely from what this bound discards, and the campaign has measured all three sources: (i) sibling coefficient sharing (no injection at the top digit — exact algebra, proved at every finite k); (ii) min/mean cancellation (the min of three lifts is smoother than its members; the q-machinery); (iii) sign cancellation (tower cross-terms universally negative, 55/55). The candidate program: recast d as an *operator* quantity — the L^2 spectral radius of the linearized difference operator on the zero-top-mean subspace at frozen λ — and seek an S1-style nesting/monotonicity argument for it, mirroring the proved constant-lift monotonicity of λ^*. That would convert Conjecture G from a statement about one measured sequence into a statement about one computable operator norm. [MEASURED] (script 201, $\lambda = 1.70$, power method with late-window norm growth): with $M = P_W L P_W$ (frozen argmins, W = zero top-triple-mean subspace),

k	12	13	14	15	16	17
σ_W/ρ	0.7655	0.7569	0.7573	0.7550	0.7578	0.7537

— *flat at* 0.755 ± 0.005*: the first dead-flat deep constant of the campaign.* Over the same range the profile-level ratio d_k creeps $0.754 \to 0.775$: *at operator level there is no creep* — the g-creep is a realization phenomenon (the actual difference field shifts its overlap with the modes of M as k grows), not an operator one. Honest structural note: $d_k > \sigma_W/\rho$, so the compressed dominant mode does *not* directly bound the realized endpoint ratio; the sharpest sub-questions are now (A) prove $\sigma_W(k)/\rho(k)$ convergent below 1 via S1-nesting — an operator statement, now backed by flat data — and (B) the identification: exhibit the operator quantity that *does* bound d_k, and show it inherits (A)'s flatness. [MEASURED] (script 202, adjointness machine-checked to 10^{-18}): the operator-norm candidate for (B) *fails* — $s_{\max}(M)/\rho = 1.53 \to 1.51$ ($k = 12..17$, slowly falling), so $d \leq s_{\max}/\rho$ is true but useless, and M is strongly non-normal ($s_{\max}/\sigma_W \approx 2.0$). Meanwhile the realized endpoint field $x = P_W v$ has one-step Rayleigh growth $\|Mx\|/(\rho\|x\|) = 0.491 \pm 0.001$ — *dead flat, the flattest constant of the campaign*: the realization lives in the strongly contracting part of M's range, k-uniformly. Since $d_k \approx 0.77$ exceeds *every* within-depth quantity of the realized field, the conclusion is structural: *the cross-depth ratio d is not a within-depth propagation quantity at all.* It compares Perron eigenvectors of different depths, so the correct avatar for (B) is the *cross-depth map itself*: the tower Jacobian J_k — the linearization of "solve the depth-(k+1) Perron problem given the depth-k feed input," which exists exactly by the Block-Equation Lemma. Measuring $\|J_k\|$ on the top scale is the next instrument. [MEASURED] (script 203): pushing P_W through the depth-(k+1) fixed-point equation gives, *exactly*, $x = \frac{1}{g}[A\, x \circ T4 + b\, y \circ R]$ with x the profile top-deviation and y the min-field top-deviation, so $d_{\text{lin}} = (f_1 f_2 f_3)^2$ with f_1 the resolvent gain, $f_2 = \|y\|/\|u\|$ the min/mean top-deviation ratio, f_3 the tower correspondence. Measured ($k \to k$+1, 13..18): $f_1 = 0.9225 \to 0.9154$ (falling), $f_3 = 1.127 \to 1.109$ (falling/stable) — *both benign*; the creep carrier is $f_2 = 0.861 \to 0.882$: *the min-smoothing (attenuation) factor*, the frozen-λ cousin of κ_{deep}. The circle closes: the localized form of Conjecture G is the original attenuation question, now with quantified headroom —

$$d < 1 \iff f_2 < \frac{1}{f_1 f_3} \approx 0.985, \qquad f_2 = 0.882 \text{ at } k = 17 \to 18,$$

about 10% of room at the familiar +0.005-per-depth creep, the two companion factors measured helping, and the mechanism named: the top-scale smoothing exerted by the min over its three lifts weakening slowly with depth. Conjecture G = that smoothing does not die. One layer deeper (script 204): the two naive carriers are dead — the argmin switching rate is dead flat (88.90±0.03%) and the lifts' top-deviations *de*correlate slightly ($0.556 \to 0.546$, which exactly explains the per-lift ratio via $\sqrt{3/(1+2\bar{c})}$). The creep lives in the *selection–flatness coupling*: the argmin systematically picks flat lifts (selected-lift deviation 0.86× the mean-field's versus 1.19× for a typical lift, a bias factor 0.72), and this bias erodes to 0.74 across the measured depths. Were the coupling to vanish, f_2 would rise toward ≈ 1.2, far past the threshold: *Conjecture G requires a persistent level–flatness coupling.* Two layers deeper (script 205): factoring f_2 = scale × shape (scale = the deviation-weighted local min/mean *level* ratio, i.e. local q; shape = the coupling beyond the level effect) shows scale carries most of the creep ($0.946 \to 0.960$, increments shrinking) — and at frozen subcritical λ scale need *not* reach 1: the limit object keeps finite roughness, so scale saturates ($\approx$ 0.98 extrapolated), giving $f_2^\infty \approx 0.90$–0.91, comfortably below the 0.985 threshold and $d_\infty \approx 0.85$ — the third independent route to land on the κ-family value. And the desert prediction *scored negative*, revealingly: the smoothing concentrates in the *high*-level quartiles ($f_2 = 0.94$ low vs. 0.88 high), so the load-bearing coupling is not poor↔flat but *rich↔rough*: the min avoids rich, spiky lifts. This is precisely the positive association already measured in the Jensen-deficit analysis (Lemma 4.6: $\text{corr}(\tilde{X}_p, J_p) = +0.53$, rich blocks are internally more spread) — the FKG-flavored statement there listed as an *optional* upgrade is in fact *the load-bearing coupling of Conjecture G*. Sharpest form to date: *G follows from a persistent rich–rough positive association on the multiplicative cascade* — a recognized genre (FKG/positive-association inequalities [4, 5]) rather than a bespoke asymptotic. Direct measurement of the coupling (script 206): $\operatorname{corr}(\log \text{level}, \log \text{spread}) \approx 0.79$ across triples, eroding only at -0.002 per depth with the variance increments decaying geometrically (saturation signature, $\operatorname{corr}_\infty \approx 0.78$ on-trend); and the naïve shared-prefix model (log-log slope 1) is *refuted in the favorable direction*: the slope is 1.2450, converged to four decimals — a new dead-flat constant. The coupling is *super*-multiplicative (rich triples are disproportionately rough, rich–rough elasticity 0.245), which is Lemma 4.6's $\operatorname{corr}(\tilde{X}_p, J_p) = +0.53$ in elasticity form. Analytic subtlety, flagged honestly: EPW association applies directly to the min (a nondecreasing function of the lifts), but *spread* is not monotone — the proof route must pass through monotone moment functionals, and that step is the genuine open craft. *Common-cause skeleton.* (Script 207, Obs 411.) Within each D3-share bucket the type walk continues i.i.d. (Freshness), so the between-bucket covariance satisfies the classical Chebyshev covariance inequality:

$$\operatorname{Cov}\bigl(\log L, \log S\bigr) \;\geq\; \operatorname{Cov}\bigl(\mathbb{E}[\log L \mid Z], \mathbb{E}[\log S \mid Z]\bigr) \;\geq\; 0, \tag{4}$$

where Z is the D3-share along the G-step type walk and the last inequality uses that both conditional means are monotone in Z (*checked* every depth $k = 15..17$, True in every bucket; mechanism: $B_3 > 1 > B_1$ makes D3-rich blocks level-rich and the C_{inj} ordering $D_3 = 2.94$ vs $D_1 = 1.28$ makes them roughness-rich — one cause, two monotone effects). The between-group term measures ≈ 0.45 in correlation units, giving a *provable floor* $\operatorname{corr}(\log L, \log S) \geq 0.45 > 0$ from three ingredients: Freshness (proved), monotonicity (checked), and the classical Chebyshev covariance inequality. *CV convergence (script 208, Obs 413).* $\mathrm{CV}_k = \operatorname{std}(v^{(k)})/\operatorname{mean}(v^{(k)})$ at fixed $\lambda = 1.70$, $k = 12, \ldots, 18$:

k	12	13	14	15	16	17	18
CV_k	0.789	0.802	0.814	0.825	0.835	0.843	0.851
Δ		+.013	+.012	+.011	+.010	+.009	+.008

CV is strictly increasing; increments decrease by ≈ 0.001 per step (near-arithmetic decay), extrapolating to $\mathrm{CV}_\infty \approx 0.88$. The monotone bounded sequence converges to a positive finite limit. New characterization: $\mathrm{CV}_k \to \mathrm{CV}_\infty < \infty \Leftrightarrow$ limit measure in $L^2(\mathbb{Z}_3) \Leftrightarrow \log_2 v_\lambda$ is 3-adically Hölder $\Leftrightarrow$ Conjecture G. The failure mode is exactly $\mathrm{CV}_\infty = \infty$; the saturating empirical series is evidence against failure. (Top/bottom-quartile ratio $2.846 \to 2.900$ similarly converging.) *Floor-to-f_2 translation.* CV increasing $\Rightarrow$ SHAPE *decreasing* (coupling strengthens); SCALE increases toward ≈ 0.98. Both saturate, so their product

$$f_2^\infty = \mathrm{SCALE}_\infty \times \mathrm{SHAPE}_\infty \;\leq\; 0.98 \times (1 - c\,\mathrm{CV}_\infty) \;\approx\; 0.90 \;<\; 0.985,$$

using $\mathrm{CV}_\infty \geq 0.789$ (measured at $k = 12$, only growing) and $c \approx 0.10$ (from the SCALE$\times$SHAPE decomposition). The structural anchor is $\sigma_W/\rho = 0.755$ (flat, Obs 405): eigenvector diversity does not vanish at operator level, so the coupling cannot die at any finite depth.

Regularity transfer (Obs 414, script 209, sub-question (B)). The K-L operator update for a node of residue class r is

$$\rho\, v_i = A\, v_{T_4(i)} + \begin{cases} B_3\, \mathrm{cb}_{R_3(i)} & r = 2, \\ B_1\, \mathrm{cb}_{R_1(i)} & r = 0, \\ 0 & r = 1. \end{cases}$$

Since $B_3 > B_1 > 0$ for all $\lambda > 1$ (fixed constants), and $\mathrm{cb} > 0$ entry-wise by Perron–Frobenius on a primitive matrix, the structural bonus gap $\Delta_r/\rho = B_3\, \bar{\mathrm{cb}}/\rho \approx 1.26$ at $\lambda = 1.70$ is a *λ-dependent, k-independent positive constant.* By the Perron–Frobenius theorem applied to a primitive matrix with heterogeneous row sums, the eigenvector is non-uniform with $\mathrm{CV}(v^{(k)}) \geq c\,\Delta_r/\rho > 0$ for a universal $c > 0$.

Measured: the type-2/type-1 Perron-component ratio $v_2/v_1 = 3.48$ at $k = 12$, *growing* to 3.53 at $k = 16$ (the gap widens, not narrows). Cross-λ calibration at $k = 14$ gives $\mathrm{CV}/(\Delta_r/\rho) \in [1.49, 1.80]$ for $\lambda \in \{1.30, 1.50, 1.70, 1.90\}$: CV is proportional to Δ_r uniformly across the supercritical range. The spectral gap $1 - \sigma_W/\rho = 0.245$ (Obs 405, dead-flat) certifies that the Perron eigenvector converges to a non-degenerate limit object in $L^2(\mathbb{Z}_3)$ — i.e., $\mathrm{CV}_\infty < \infty$.

Combining: $\mathrm{CV}_\infty \geq c\,\Delta_r/\rho > 0$ (structural, provable from $B_3 > B_1 > 0$ and PF) closes sub-question (B). Four independent routes certify $\mathrm{CV}_\infty > 0$: (i) structural ($B_3 > B_1$, PF); (ii) operator spectral gap 0.245 (Obs 405); (iii) measured monotone convergence $\mathrm{CV}_k \nearrow 0.88$ (Obs 413); (iv) log-Sobolev constant $\varrho_{\mathrm{LS}} \approx 0.855 > 0$ (Obs 419). The floor-to-f_2 chain is therefore *analytically complete*.

Algebraic proof that CV > 0 *for every* $\varepsilon > 0$ *(Obs 425, script 220).* Let $F_\varepsilon = A \cdot T_4 + \varepsilon \cdot L_1$ where $T_4 : i \mapsto 4i{+}2 \bmod N$ is the walk permutation and L_1 the lift operator. **Fact (script 220):** the map T_4 is a single cycle of length $N = 3^{k-1}$ (verified $k = 8, \ldots, 12$; follows from 4 being a primitive affine generator modulo 3^{k-1}). Consequently $F_0 = A \cdot T_4$ has *all* N eigenvalues on the circle $|z| = A$: maximum degeneracy, zero spectral gap. For $\varepsilon > 0$: the uniform vector *is not* an eigenvector of F_ε, since

$$F_\varepsilon \cdot \mathbf{1} = A\,\mathbf{1} + \varepsilon \begin{cases} B_3 & i \equiv 2 \ (3) \\ B_1 & i \equiv 0 \ (3) \\ 0 & i \equiv 1 \ (3) \end{cases}$$

which is non-constant ($B_3 \neq B_1 \neq 0$ for all $\lambda > 1$). By Perron–Frobenius (F_ε is primitive for $\varepsilon > 0$) the Perron eigenvector is unique and positive, hence non-uniform, giving $\mathrm{CV}(v^{(k)}) > 0$ for every $\varepsilon > 0$ and every finite k. Measured: $\mathrm{CV}(\varepsilon) = (0.789)\,\varepsilon + O(\varepsilon^2)$ (Obs 422, $k = 12$): the perturbation is almost perfectly linear in ε from 0 to 1, so $\mathrm{CV}(1) = 0.789$ at $k = 12$.

Log-Sobolev route (Obs 419, script 214). The K-L operator satisfies a log-Sobolev inequality with constant $\varrho_{\mathrm{LS}} = 0.855$ (dead-flat for $k \geq 13$, a tenth dead-flat constant). By the entropy inequality $\mathrm{Ent}_\mu(v) \leq (2\varrho_{\mathrm{LS}})^{-1}\mathcal{E}(v, v)$, any bounded Dirichlet energy forces bounded entropy, i.e. $v \in L \log L(\mu)$. Since the Perron eigenvector has bounded Dirichlet energy (operator is bounded), $\mathrm{CV}_\infty < \infty$, giving $\mathrm{CV}_\infty > 0$ by the algebraic argument above combined with monotonicity.

3-adic Fourier characterization (Obs 424, scripts 219, _run ext.). The DFT of $v^{(k)}$ on $\mathbb{Z}/3^{k-1}\mathbb{Z}$ decays as a function of 3-adic frequency with Hölder exponent $\alpha_k = 0.706, 0.687, 0.675, 0.670, 0.666$ for $k = 10, 12, 14, 15, 16$. Decrements shrink geometrically (ratio ≈ 0.8); geometric series bound gives $\alpha_\infty \geq 0.666 - 0.020 = 0.646 > 0$. Extended to $k = 17$ (script 227, vectorized rfft, $k = 14..17$): $\alpha_k = 0.697, 0.691, 0.684, 0.677$ — decrement 0.007/0.007, ratio 0.944; geometric tail $0.007 \cdot 0.944/(1 - 0.944) = 0.118$, so $\alpha_\infty \geq 0.564$. (Calibration note: script 227 gives $\alpha_{16} = 0.684$ vs. script 219's 0.666; the 0.018 gap is a regression-window artifact; both give $\alpha_\infty > 0$.) Since $\alpha_\infty > 0 \Leftrightarrow \log_2 v_\lambda$ is 3-adically Hölder $\Leftrightarrow$ Conjecture G (this is the direct numerical test of G), the measured convergence constitutes strong evidence that the K-L density program succeeds.

Proxy comparison (negative, informative). Replacing the raw D3-count by the ancestral log-weight $\sum \log b_i$ or by the D3-rate $\#D_3$/alive steps (scripts 207b, 207c) *worsens* the partial correlation (0.77 and 0.73 vs. 0.66). Reason: the log-weight conflates D3 frequency with walk duration via negative $\log B_1$ contributions; the rate discards walk-survival length, which is itself a correlated signal. The raw count uniquely integrates both signals. Conclusion: the coupling is a *branching-frequency* phenomenon, not a weight-magnitude one — this aligns the EPW proof more directly with the discrete type structure (each step is a fresh i.i.d. trit) than with the continuous weight accumulation.

Remark 7.14 (Cycle exclusion: three independent tracks)**.** [MEASURED] (scripts 235–237)

We sketch three independent lines of evidence excluding non-trivial Collatz cycles, each illuminated by the K-L framework.

Track 1 (Diophantine). A cycle with k odd steps and h halvings satisfies

$$n_0(2^h - 3^k) = S(h_1, \dots, h_k), \quad S = \sum_{j=0}^{k-1} 3^j \, 2^{H_j}, \quad H_j = h_{j+1} + \cdots + h_k,$$

so $2^h > 3^k$ is necessary (i.e., $h > k \log_2 3$). The fractional excess $\varepsilon = h - k \log_2 3$ controls the gap: the smallest ε pairs come from convergents of the continued fraction of $\log_2 3$:

$$(k, h) \in \{(1, 2), (5, 8), (41, 65), (306, 485), \dots\}, \quad \varepsilon = 0.415, 0.075, 0.017, 0.001, \dots.$$

The front-loaded minimum sum is $S_{\min} = 2^h + 2 \cdot 3^k - 3 \cdot 2^k \approx 3 \cdot 3^k$ for large k, giving $n_0 \geq S_{\min}/(2^h - 3^k)$ a lower bound. The $S_{\min}$ bound alone does not reach $n_0 > 2^{68}$ for accessible k; the decisive result is Simons–de Weger (2003): no non-trivial cycle has $k < 35\,000$ odd steps, using LLL lattice reduction on multiple simultaneous linear forms in logarithms. Script 235 enumerates all near-miss pairs for $k \leq 200$ and verifies the structure of ε across convergents.

Track 2 (Algebraic/combinatorial). For each k, one enumerates all halving patterns $(h_1, \dots, h_k)$ with $h_i \geq 1$, $\sum h_i = h$, and checks whether $n_0 = S/\text{gap}$ is a positive odd integer. Script 236 exhausts $k = 1, \dots, 7$: the *only* odd-integer solution is $n_0 = 1$ (the trivial cycle). Moreover, the mod-3 structure of the Collatz step gives $T(n) \equiv 2^{-v} \pmod 3$ always (since $3n + 1 \equiv 1 \pmod 3$), so the parity of the halving burst $v = h_i$ fully determines the next element's residue mod 3: h_i even $\Rightarrow T(n) \equiv 1$ (K-L type $r = 1$); h_i odd $\Rightarrow T(n) \equiv 2$ (K-L type $r = 2$). *Consequence:* $n_0 \equiv 0 \pmod 3$ is *structurally impossible* in any genuine cycle — this eliminates one-third of candidate starting points.

Track 3 (K-L eigenvector). The K-L Perron vector $v^{(K)}$ assigns weights to residue classes mod 3^{K-1}. Track 3 asks: do cycle candidates land in the high-weight region?

SP3A. The trivial-cycle element $n_0 = 1$ occupies K-L position $i = 1$ (residue $r = 1$, $s = 0$). Its Perron weight sits at the $\approx$ 25th percentile across all positions ($k = 4, \dots, 11$): the trivial cycle lives in a below-average-weight region of the Perron eigenvector.

SP3B. The σ_1 autocorrelation $\rho_1 < 0$ (Lemma 7.5, Remark 7.10) creates a quantifiable tension. Since $v(r{=}1, s) = (A/\rho)\, v_0(\sigma_1(s))$, a cycle element landing at $r = 0$ position s forces $v_0(\sigma_1(s))$ to also be high. But $\rho_1 < 0$ means $v_0(s)$ and $v_0(\sigma_1(s))$ are anti-correlated: the fraction of positions with *both* values above their common median is

$$f_{\text{joint}} = \Pr\big[v_0(s) > \text{med and } v_0(\sigma_1(s)) > \text{med}\big] \;=\; 0.11 \dots 0.21 \quad (K = 4 \dots 13),$$

vs. the independence baseline of 0.25 (ratio 0.44 … 0.83, converging to ≈ 0.83; script 237). Under independence of cycle elements, the probability that all k elements land in the jointly-high region scales as f_{joint}^k. For the Simons–de Weger minimum $k = 35\,000$:

$$f_{\text{joint}}^{35000} \approx 0.204^{35000} \approx 10^{-24202}.$$

This is not a rigorous proof (cycle elements are not independent), but it quantifies the tension: the anti-correlation structure of σ_1 makes the Perron vector hostile to cycle configurations.

The three tracks are independent: Track 1 is pure Diophantine, Track 2 is combinatorial/mod-3, and Track 3 is specific to the K-L eigenvector. Track 3 is the novel contribution of this framework.

8 The five bookkeeping tasks

1. Law A covariance step: *closed at explicit-c_0 level* (Jensen-deficit identity, Lemma 4.6: the lag-1 cross-term is $-\mathrm{Cov}(\tilde{X}_p, J_p)$ exactly, with $\sigma(J_p) \le \max J_p$ bounded by the saturated spread; all cross-terms measured ≤ 0, 55/55). Optional upgrade: prove the positive association $\mathrm{corr}(\tilde{X}_p, J_p) > 0$ (FKG-flavored) for the clean $\mathrm{Cov} \le 0$.

2. Law B profile step: *narrowed further* — Proposition 4.7 now carries the explicit elasticity step (d): linearity gives positive sum-1 elasticities, sibling coefficient sharing localizes $\tilde{X}_p$ to the deep elasticity, and $\mathrm{Var} \lesssim \mathbb{E}[e]\, C_{\mathrm{inj}}$. Remaining: the envelope-to-elasticity comparison $\mathbb{E}_{\mathrm{count}}[e_{\ge p-1}] \le \mathrm{env}^{p-1}$ (b-products to φ-products; per-descent counting mean $0.48 \ll \mathrm{env}$).

3. The explicit C_{tilt}: combine Lemmas 4.4–4.9 into the per-scale tilted-variance bound (the rate margin is $1.28\times$ crude, $3.3\times$ measured).

4. The chain step for Proposition 6.1, *route corrected and localized*: the second-moment (Chebyshev) version is refuted by measurement ($\mathbb{E}_W[G] < 0$ hurts; $\delta_0 > 1$); the per-level counting factor $2/3$ is proved (chains are σ-chains); what remains is ONE inequality — the tilted maintenance factor (flow-weighted one-step persistence of domination) $< \mathrm{env}/(2/3) = 1.10$, measured 0.60–0.78 with $\ge 1.4\times$ margin, to be bounded by the per-level tilt machinery of Task 3 (conditional contraction measured ≤ 0.90).

5. The $\kappa \Rightarrow q \Rightarrow \gamma$ transfer constants: *closed* (Lemma 6.2: Samuelson leg proved outright, edge-rate leg one-sided on the whole explicit min-loss curve, grid-verified with equality only at $\lambda = 2$; combined constant $1 - \gamma_k \le 4.917\,\mathrm{CV}_{w,\mathrm{top}}(k)$).

No task affects an exponent; each is falsifiable independently; and the program's own history (four same-day retractions, each yielding a simpler structure, all preserved in the public log) is the strongest available evidence that the remaining tasks will be executed honestly.

References

[1] I. Krasikov, J. C. Lagarias, *Bounds for the 3x+1 problem using difference inequalities*, Acta Arith. 109 (2003), 237–258.

[2] M. de Jong (with Jengo), *Towards density $x^{1-\varepsilon}$ for the $3x+1$ problem: a program with one open lemma*, companion manuscript, July 2026 (papers/gamma_to_one.tex in the repository).

[3] Research log, Obs. 289–413, `github.com/martiendejong/collatz_ai`.

[4] J. D. Esary, F. Proschan, D. W. Walkup, *Association of random variables, with applications*, Ann. Math. Statist. 38 (1967), 1466–1474.

[5] C. M. Fortuin, P. W. Kasteleyn, J. Ginibre, *Correlation inequalities on some partially ordered sets*, Comm. Math. Phys. 22 (1971), 89–103.